

Final Report

P4B: Solar Mini-Grid: Socioeconomic Impact Analysis

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Abstract— In early 2024, Renewvia published a socioeconomic impact analysis for mini-grids in rural Africa. The conclusions drawn from the study are broad and general. Using the open-source data from the study, we analyzed the data to understand the impact of Renewvia's work with solar mini-grids on local communities.

Keywords—solar mini-grid, socioeconomic impact, education

I. BACKGROUND INFORMATION

Energy is essential to maintaining daily life and is a fundamental driver of socioeconomic development. Among energy sources, electricity plays a critical role in improving quality of life and supporting social and economic progress. However, in rural areas, electricity supply is often difficult to access, creating a persistent challenge that lowers both quality of life and socioeconomic development. Especially for Sub-Saharan Africa, where faces a significant energy deficit, with only 45% of the population having access to electricity. The majority of those without access live in rural areas due to several factors: including lack of financial ability to extend the main power grid, low population density, making traditional grid extension not cost-effective, and social factors that limit the adoption of electrification projects.[1]

As a result, the lack of reliable electricity hinders progress in critical areas such as education, healthcare, and income generation, perpetuating poverty and limiting development opportunities. Bridging this energy gap is vital to empower these underserved regions and enable them to thrive

Solar mini-grids offer a promising solution by providing decentralized, reliable, and sustainable energy systems tailored to the needs of rural communities. These systems not only enhance access to electricity but also act as catalysts for broader social and economic transformation. For example, they have

demonstrated the potential to improve education, healthcare, and economic productivity in off-grid areas.

However, challenges remain in fully understanding and maximizing their impact, particularly regarding subgroup-specific outcomes such as gender equity in education and household-level socioeconomic changes

In early 2024, Carabajal et al. Published an analysis of the socioeconomic impact on solar mini-grids in rural African communities. The analysis focuses on rural households, schools, businesses, and healthcare facilities in Kenya and Nigeria, which are directly impacted by the mini-grid installations. The study analyzed data from many households and businesses, revealing that their quality of life has improved in many ways. However, the study is extensive, and there is a need for subgroup analysis and variable correlation. Using the data used in this study, the investigations are performed to reveal the correlation between education and solar mini-grids in rural communities. [1]

Current analysis has achieved significant progress in revealing how mini-grid projects improve socioeconomic outcomes in rural areas. In education, mini-grids have increased school enrollment rates, extended study hours, and enhanced academic performance, with 46% of households reporting positive changes in children's education and 92% of schools observing better outcomes due to reliable electricity. In healthcare, electrified clinics reported a 47% increase in refrigeration availability for vaccines and medicines, substantially improving service delivery and healthcare outcomes. Economically, businesses expanded operational hours, increased revenue, and contributed to greater household income stability, highlighting the transformative potential of solar mini-grids.

Despite these achievements, the approach still lacks detailed subgroup analyses, particularly in understanding gender equity

and indirect educational benefits. Additionally, there is insufficient exploration of nuanced socioeconomic relationships, including how electrification indirectly impacts household roles and time allocation. These gaps are compounded by infrastructure limitations, as extending centralized grids to rural areas remains cost-prohibitive due to low population density and logistical challenges. Ensuring scalability and sustainability also requires robust community engagement and effective management systems. Furthermore, existing studies often lack comprehensive data and detailed analyses to capture the full spectrum of social impacts, such as changes in household dynamics and educational outcomes.

Our project employs advanced analytical techniques and a focus on subgroup exploration to overcome these challenges. By analyzing open-source data from stakeholders, it identifies key variables such as home lighting, economic activity, and household stability that significantly influence educational outcomes. The project also prioritizes subgroup analysis to understand specific impacts, including how energy access benefits both boys and girls while reducing gender disparities. Beyond immediate benefits, the project adopts a holistic approach to evaluate how solar mini-grids indirectly affect healthcare, economic productivity, and overall community well-being, providing a comprehensive framework for understanding and maximizing the impacts of decentralized energy solutions.

II. MOTIVATION OF SOLUTION

The mini-grid project is not only designed to enhance access to reliable and affordable energy sources but also aims to transform local societies into more modernized and sustainable communities. Exploring the intricate relationships between these energy projects and other key aspects of socioeconomic development presents a compelling area for discussion and analysis.

The above background information motivates the solution to focus on education by highlighting the transformative potential of solar mini-grids in addressing critical gaps in rural electrification and its direct and indirect impacts on socioeconomic development. The statement identifies education as one of the key areas where energy access can significantly improve quality of life. Mini-grids have already demonstrated their ability to increase school enrollment rates, extend study hours, and enhance academic performance.

Education represents a measurable and impactful domain through which the broader socioeconomic advantages of solar mini-grids can be quantified. By focusing on those areas, the analysis can offer targeted insights into how renewable energy solutions foster human capital development, a critical driver of long-term socioeconomic progress.

Therefore, the need for detailed subgroup analyses and variable correlations, particularly concerning gender equity and indirect educational benefits, presents an opportunity to deepen understanding. The background emphasizes gaps in exploring how electrification impacts household dynamics, such as reducing time burdens on children for household chores, which disproportionately affect girls. Addressing these gaps aligns with the project's goal to use advanced analytical techniques to

examine key variables like home lighting and household stability, which are pivotal for educational outcomes.

The motivation stems from the dual recognition of existing achievements and limitations. While solar mini-grids are proven to enhance socioeconomic stability, the lack of nuanced insights into subgroup-specific impacts—such as gender disparities in education—drives the project's focus. By employing a holistic approach that captures both direct and indirect benefits of mini-grids on education, the project aims to create actionable strategies for scaling these energy solutions and ensuring equitable and sustainable development in rural communities.

By understanding these impacts, we can not only highlight the immediate benefits of electrification but also provide valuable insights into its potential to drive sustained development and improve quality of life in underserved communities.

III. INFLUENCE ON EVALUATION AND DEPLOYMENT

A. Evaluation Influences

The background information emphasized the critical role of electricity in driving socioeconomic development, particularly in education, healthcare, and economic productivity. This guided the evaluation strategy by identifying measurable outcomes:

- **Education:** Evaluation focused on metrics like school enrollment rates, extended study hours, and academic performance. Data-driven methodologies, such as Recursive Feature Elimination (RFE) and regression models, were employed to pinpoint significant predictors of educational improvements, including home lighting and business activity.
- **Healthcare:** Improved access to refrigeration and powered medical equipment in clinics served as key indicators of healthcare advancements.
- **Economic Productivity:** Metrics such as household income stability and increased business hours captured the economic benefits of electrification.

The sustainability analysis further underscored the importance of subgroup analyses, especially regarding gender equity and household dynamics. This led to a focused evaluation on how mini-grids impacted boys' and girls' education differently and how shifts in household labor roles affected overall time allocation. The analysis also leveraged machine learning to identify and analyze significant variables influencing these outcomes, ensuring a data-driven approach to evaluation.

B. Deployment Influences

Deployment strategies were shaped by the challenges and opportunities highlighted in the earlier sections. Key influences included:

- **Targeted Implementation:** The identification of rural areas as high-need zones guided the prioritization of underserved communities in Kenya and Nigeria. The modular and scalable nature of solar mini-grids was particularly suited to addressing the logistical and

financial constraints of deploying energy solutions in low-density regions.

- **Community-Centric Design:** Sustainability insights emphasized the importance of community engagement. Deployment plans included training local technicians for system maintenance and incorporating community input during the planning phase to ensure cultural and operational alignment.
- **Affordability and Accessibility:** Strategies such as tiered pricing models and pay-as-you-go systems were implemented to balance inclusivity with financial sustainability, allowing low-income households to access electricity while maintaining revenue streams for operational costs.
- **Environmental Integration:** Deployment incorporated renewable energy technologies, reducing reliance on diesel generators and mitigating greenhouse gas emissions. This not only supported environmental sustainability but also improved public health by reducing air pollution.

In conclusion, the integration of evaluation metrics and deployment strategies derived from Background Information ensures that the solar mini-grids not only meet immediate energy needs but also serve as catalysts for sustainable development.

IV. SUSTAINABILITY ANALYSIS

Our project focused on data analysis and deriving text insights from a previous study conducted by our partner organization, culminating in a journal letter publication. This project is designed to be highly sustainable for our partner, as it requires no future maintenance costs or external support. At project completion, we will provide our partner with all data and research in a readily accessible format, including image and text files, ensuring that they can independently manage and reference this information going forward.

The sustainability of solar mini-grids as a transformative solution in rural Kenya and Nigeria hinges on their capacity to address socioeconomic challenges while ensuring operational resilience and financial viability over the long term. This analysis evaluates sustainability through three interconnected dimensions: economic sustainability, social sustainability, and operational sustainability. By examining these dimensions, the analysis provides a holistic understanding of how solar mini-grids can contribute to sustained rural development.

A. Economic sustainability

Economic sustainability is a cornerstone of mini-grid success, as it ensures that the systems remain financially viable for operators and affordable for users. Key insights include:

- **Boosting Local Economies:** The presence of mini-grids has led to increased business activity, as evidenced by the strong association between recent business operations and improved educational outcomes (*Chi-square statistic: 299.94, p-value: 0.000*). Households with active businesses benefit from enhanced income streams, which support both energy payments and broader community development.

- **Energy Affordability:** Affordability remains a critical factor in sustaining energy access. Households that experienced improved school performance reported higher energy payments on average (*t-statistic: 5.15, p-value: 0.000*), suggesting that households prioritize energy for its developmental benefits.

- **Catalyzing Broader Investments:** Mini-grids foster conditions for further investment in rural areas, such as in agricultural processing and small-scale manufacturing, by providing reliable electricity. This multiplier effect helps anchor long-term economic growth in these communities.[2]

B. Social Sustainability

The social sustainability of mini-grids is evident in their ability to enhance quality of life and promote equitable access to resources. Key areas of impact include:

- **Educational Improvements:** Reliable electricity has contributed to significant advancements in education. Home exterior lighting (*Chi-square statistic: 119.73, p-value: 0.000*) and community lights (*Chi-square statistic: 104.35, p-value: 0.000*) were strongly associated with improved educational outcomes. These improvements are attributed to safer environments for study and extended learning hours, highlighting the transformative potential of stable energy access.
- **Gender Equity and Empowerment:** Mini-grids help reduce gender disparities by alleviating domestic burdens often placed on women and girls, such as cooking and water collection. These changes allow for greater participation in education and income-generating activities, empowering women and fostering gender-balanced community development.[3]
- **Enhanced Healthcare Services:** Electrified clinics reported a 47% increase in refrigeration availability for vaccines and medicines, significantly improving health outcomes. With better-equipped facilities, communities experience reduced healthcare risks and higher service reliability, which strengthens overall well-being.[1]
- **Environmental Benefits:** Transitioning from diesel generators to solar-powered mini-grids reduces greenhouse gas emissions and air pollution, contributing to environmental sustainability and improved public health.[4]

C. Operational Sustainability

The operational sustainability of mini-grids is critical to ensuring their long-term viability and scalability. Factors influencing this dimension include:

- **Community Engagement and Ownership:** Community involvement in managing and maintaining mini-grid systems enhances local ownership and operational reliability. Engagement initiatives, such as training community members in system maintenance, ensure the

longevity of the infrastructure while fostering a sense of collective responsibility.

- **Scalability and Modular Design:** The modular nature of solar mini-grids makes them an ideal solution for low-density rural areas, where traditional grid extension is cost-prohibitive. By scaling incrementally to meet growing demand, mini-grids can adapt to community needs while minimizing upfront investment risks.[4]
- **Technical Reliability and Resilience:** Ensuring the technical reliability of mini-grids involves investing in durable materials, advanced battery technologies, and weather-resistant designs. These enhancements help mitigate seasonal fluctuations in energy supply and demand, ensuring consistent service delivery.[5]
- **Scalability and Flexibility:** The modular nature of solar mini-grids allows for scalable deployment, making them suitable for remote and low-density populations where traditional grid extension is not feasible.[6]
- **Data-Driven Management:** Leveraging machine learning models, as demonstrated in this project, can optimize system performance by identifying key drivers of energy consumption and household impacts. Insights from predictive analytics can guide maintenance schedules, anticipate system failures, and inform expansion plans

V. LESSONS LEARNED

Throughout the course of this project, we gained several key insights that will be invaluable for future projects. These lessons are detailed below:

A. Planning and Multitasking

- **Time Management:** Early in the project, we realized the importance of checking ahead on future assignments and beginning multiple assignments simultaneously. Due to time constraints and resource availability, some tasks required significant lead time to complete. For instance, one assignment required nearly a month to finalize because it involved scheduling meetings with stakeholders, conducting surveys, and subsequently creating a comprehensive report based on the findings. Proper planning and proactive multitasking were essential in meeting deadlines and ensuring the quality of our work.
- **Preemptive Resource Allocation:** Anticipating resource needs for data processing and statistical analysis enabled smooth project execution without delays. For example, the appropriate machine learning approach may need several rounds to be confirmed, allocating bandwidth between team mates to move forward in parallel is key to making sure we don't miss key check points.

B. Data Utilization

- **Importance of Data Quality:** A thorough understanding of the data's characteristics was essential for conducting meaningful analyses. Preprocessing techniques, such as feature elimination and variable transformation, ensured

robust model performance. Since the raw data comes from survey, lots of data cleaning work needed to be done, choosing appropriate cleaning methods is important to keep the data quality.

- **Advanced Analytical Techniques:** Leveraging machine learning models like Random Forest and Recursive Feature Elimination (RFE) provided deeper insights into key factors influencing socioeconomic outcomes. Advanced techniques can help to reduce workload in feature selection.
- **Correlation vs. Causation:** Distinguishing between correlated and causal relationships was a critical consideration in interpreting results, especially when analyzing indirect impacts on education and household dynamics. In the Socioeconomic Impact Analysis project, we pay more attention to correlation rather than causation, since the causation may consist of complex composition.

C. Communication with stakeholders

- **Stakeholder Engagement:** Regular communication with community representatives and stakeholders ensured that the deployment strategy was aligned with local needs and expectations. Periodical meeting with Dr.Selby helps us to get intime instruction and clarify the needs, for example, he recommended us to focus on education instead of healthcare because he realized that education has more fruitful insights based on his experience.

D. Technology Integration

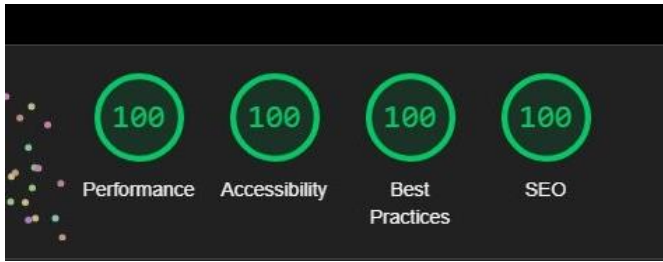
- **Data-Driven Decision-Making:** The use of predictive analytics optimized deployment and management strategies. For instance, analyzing factors like household headcount and energy payment behavior informed targeted interventions.
- **Data Gaps:** Some variables critical for subgroup analysis were either incomplete or unavailable, limiting the depth of analysis. For example, the number of kids by gender and the educational level of parents are not available in the raw data.
- **Resource Constraints:** Balancing project goals with limited resources necessitated prioritization and iterative improvements during implementation.

VI. USER AND DEVELOPER DOCUMENTATION LINKS

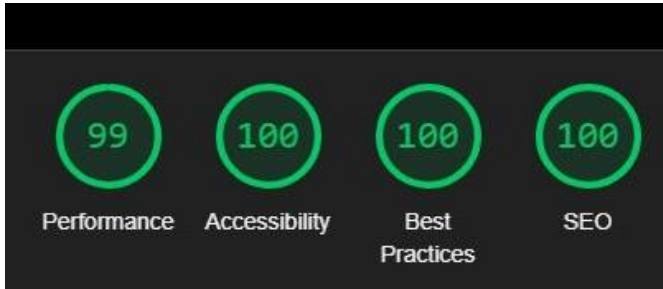
User	Documentation	Link:
		https://gtvault.sharepoint.com/:w/s/CS6150-P4B/ESiTNco4vgRNhsCb7pn2-fEByvNuFGP7mPOAlpdPxXMUYw?e=zoD76K
Developer	Documentation	Link:
		https://gtvault.sharepoint.com/:w/s/CS6150-P4B/Ebp8fnKFhyRBqNOdoEH3fCwBqn1AzcaP8s_3H1LxBd cgIw?e=2a2ehs

VII. LIGHTHOUSE METRICS & FORM FACTOR

Desktop Lighthouse Metrics:



Mobile Lighthouse Metrics:



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